

Cardiovascular and Metabolic Outcomes of Hormone Replacement Therapy in Women Aged 65 Years and Older: A Systematic Review of Randomized Clinical Trials

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Background: The cardiovascular, kidney, and metabolic effects of hormone replacement therapy (HRT) in postmenopausal women, particularly those who initiate treatment after age 65, remain incompletely characterized. The objective of this systematic review was to evaluate evidence on these outcomes, with a focus on women aged 65 years and older.

Methods: PubMed, Embase, Scopus, and the Cochrane Library were searched for studies published between January 1, 2021, and February 28, 2026. Eligible studies included randomized controlled trials and secondary or post-trial analyses evaluating HRT in postmenopausal women, particularly women aged 65 years and older. Outcomes included cardiovascular, renal, and metabolic endpoints. Risk of bias was assessed using RoB 2 where applicable, and evidence was synthesized narratively. The review protocol was registered in PROSPERO (CRD420261338310).

Results: Of 1701 records identified, 6 reports met the inclusion criteria, all derived from three parent trials: the Women's Health Initiative Hormone Therapy Trials (WHI-HT), Early versus Late Intervention Trial with Estradiol (ELITE), and Kronos Early Estrogen Prevention Study (KEEPS). HRT effects varied by regimen and timing. In WHI-HT analyses, conjugated equine estrogens (CEE), alone or with medroxyprogesterone acetate (MPA), lowered low-density lipoprotein cholesterol (LDL-C) but increased triglycerides and systolic blood pressure; CEE plus MPA had a less favorable coronary profile than CEE alone. ELITE supported less favorable vascular effects with later initiation, while KEEPS showed neutral long-term findings. No trial was designed specifically for women aged ≥ 65 years, and evidence in this age group was limited to subgroup analyses of broader postmenopausal women. Additionally, no studies included renal outcomes.

Conclusion: Future prospective studies should evaluate cardiovascular, kidney, and metabolic outcomes across HRT regimens and timing in older postmenopausal women to guide individualized treatment decisions.

Keywords: cardiovascular diseases, hormone replacement therapy, menopause, metabolic diseases, risk factors, systematic review

Introduction

Hormone replacement therapy (HRT) remains the most effective treatment for vasomotor symptoms and genitourinary syndrome of menopause. It is also effective for preventing postmenopausal bone loss and reducing fracture risk.¹ Yet, despite its established benefits for symptom control, the cardiovascular and metabolic effects of HRT remain complex and incompletely characterized, particularly in older postmenopausal women.^{2,3}

Both randomized controlled trials (RCTs) and observational studies demonstrate that HRT exerts pleiotropic effects on cardiometabolic physiology, including modulation of lipid and lipoprotein profiles (eg, reductions in low-density lipoprotein cholesterol (LDL-C) and increases in high-density lipoprotein cholesterol or HDL-C with oral estrogen),

vascular and endothelial function, coagulation pathways, blood pressure, glucose homeostasis, insulin sensitivity, and body fat distribution.^{4,5} These pathways also undergo unfavorable changes during the menopause transition and postmenopause, contributing to increasing cardiometabolic risk with aging.^{2,6} Kidney health is also closely linked to menopause and estrogen status. Ovarian senescence and the loss of estrogen's potentially renoprotective effects may contribute directly to declining kidney function and indirectly through the accumulation of cardiometabolic risk factors that predispose women to chronic kidney disease. Although estrogen may influence renal function, epidemiologic findings regarding the kidney effects of HRT remain conflicting, and the optimal use and dosing of hormone therapy among women with impaired kidney function have not been established. These uncertainties are particularly relevant to older postmenopausal women, in whom age-related kidney function decline and comorbidities such as hypertension, diabetes, and dyslipidemia are increasingly common.⁷

Much of the ongoing concern about HRT safety arises from RCTs showing that its effects differ according to regimen and patient characteristics. In the Women's Health Initiative Hormone Therapy Trials (WHI-HT), combined estrogen–progestin therapy was associated with increased risks of coronary heart disease, stroke, venous thromboembolism, pulmonary embolism, and breast cancer.⁸ By contrast, estrogen-only therapy remained associated with increased risks of stroke and venous thromboembolism but did not demonstrate the same increase in breast cancer seen with combined estrogen–progestin therapy.⁹ Subsequent age- and timing-stratified analyses resulted in the timing hypothesis, which proposes that the cardiovascular effects of HRT vary based on the interval between menopause onset and treatment initiation.^{10,11} Accordingly, the North American Menopause Society (NAMS) recommends that the benefit–risk profile of HRT is generally favorable for women younger than 60 years or within 10 years of menopause onset.¹

Consistent with this evolving evidence base, in February 2026, the US Food and Drug Administration (USFDA) approved labeling changes for six menopausal HRT products that removed boxed-warning statements concerning cardiovascular disease, breast cancer, and probable dementia, while retaining the boxed warning on endometrial cancer for systemic estrogen-alone products.¹²

Despite substantial progress in understanding HRT effects in menopausal women, important uncertainty remains for women who initiate therapy at older ages, particularly after age 65 years or more than 10 years after menopause onset.¹ This population is especially important because aging is accompanied by greater subclinical atherosclerotic burden, endothelial dysfunction,^{13–15} and longer cumulative exposure to estrogen deficiency, all of which may alter the physiologic and clinical effects of HRT. At the same time, older women remain underrepresented in contemporary evidence syntheses focused specifically on cardiometabolic outcomes, leaving clinicians with limited data to guide decisions regarding late initiation or continuation of therapy.

A focused synthesis of evidence in this population is therefore needed to improve individualized benefit–risk assessment. Although two recent narrative reviews^{4,5} summarized the broader evidence on HRT and cardiometabolic or cardiovascular health, neither systematically focused on randomized and post-trial evidence relevant to women aged 65 years and older or included renal outcomes as a predefined outcome domain. Accordingly, this systematic review aimed to identify, critically appraise, and synthesize evidence evaluating the cardiovascular-kidney-metabolic effects of HRT in postmenopausal women, with particular emphasis on women aged 65 years and older. The review also assessed the strengths and limitations of the available evidence, including risk of bias, identified important knowledge gaps, and highlighted priorities for future research, regulatory evaluation, guideline development, and personalized clinical decision-making.

Methods

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guideline.¹⁶ The protocol was registered in the International Prospective Register of Systematic Reviews (PROSPERO)(CRD420261338310).

Eligibility Criteria

Eligibility criteria were defined according to the PECO framework and study design. The population (P) of interest was postmenopausal women, with particular emphasis on women aged 65 years and older. Studies conducted in broader postmenopausal populations were also eligible if they reported subgroup, stratified, or follow-up findings relevant to

older women. The exposure/intervention (E) included HRT regimens used in postmenopausal women, including estrogen-only therapy and combined estrogen–progestogen therapy, regardless of formulation, dose, route of administration, duration of treatment, or timing of initiation. The comparator (C) included placebo, usual care, or alternative HRT regimens. The outcomes (O) of interest were cardiovascular, renal, and metabolic outcomes. Cardiovascular outcomes included major cardiovascular events and related endpoints, such as coronary heart disease, myocardial infarction, stroke, heart failure, hypertension, atrial fibrillation, venous thromboembolism, deep vein thrombosis, pulmonary embolism, and composite cardiovascular outcomes. Renal outcomes included kidney disease, chronic kidney disease, acute kidney injury, albuminuria, proteinuria, and changes in glomerular filtration rate. Metabolic outcomes included metabolic syndrome, type 2 diabetes, insulin resistance, dyslipidemia, obesity, overweight, and glycemic outcomes.

Eligible study designs included RCTs and secondary analyses of randomized trial populations, including subgroup analyses and post-trial or long-term follow-up reports. Studies were excluded if they involved premenopausal women, did not provide findings relevant to older postmenopausal women, included populations outside the review scope, evaluated interventions other than HRT, or did not report outcomes of interest. Observational designs, including prospective cohort studies, retrospective cohort studies, case-control studies, cross-sectional studies, and registry-based analyses, were excluded because HRT use in these studies may be influenced by baseline characteristics, indication for treatment, prescribing patterns, and other factors that could introduce confounding and bias the final estimates. Although the exclusion of observational studies narrowed the review to evidence arising from randomized exposure, it also prevented the review from representing the full contemporary evidence base. Additionally, reviews, editorials, commentaries, letters, protocols, preclinical studies, and conference abstracts without sufficient data were excluded. No restrictions were placed on geographic location or clinical setting, but only English-language studies were included.

Information Sources and Search Strategy

A systematic literature search was conducted in PubMed, Embase, Scopus, and the Cochrane Library, including the Cochrane Central Register of Controlled Trials (CENTRAL), to identify eligible literature. The search covered studies published from January 1, 2021, through February 28, 2026. Search strategies combined database-specific controlled vocabulary and free-text terms related to menopausal hormone therapy, including estrogen-only therapy, combined estrogen–progestogen therapy, estrogen replacement therapy, postmenopausal hormone therapy, systemic estrogen therapy, transdermal estradiol, and tibolone, together with terms for menopause and for cardiovascular, metabolic, and renal outcomes. Where available, filters for publication year, language, sex, and age were applied at the database level; otherwise, eligibility criteria were assessed during manual screening. Full search strategies for all databases are provided in the [Supplementary Material](#) and [Table S1](#).

Data Collection Process and Data Items

Data extraction was performed independently by two reviewers (TM and MFH) using a standardized data extraction form. Disagreements were resolved through discussion and consensus.

The extracted data included study characteristics such as author, publication year, study design, trial name when applicable, country or setting, and duration of follow-up; participant characteristics, including sample size, age, and menopausal status; intervention and comparator details, including HRT regimen, formulation, dose, route of administration, duration of treatment, and comparator type; and reported cardiovascular, renal, and metabolic outcomes, including outcome definitions, effect estimates, and measures of variability when available.

Risk of Bias (RoB) Assessment

RoB was assessed independently by two reviewers using the Cochrane RoB 2 tool,¹⁷ with disagreements resolved through discussion and consensus. RoB 2 was applied to assess potential bias across the relevant domains for each included study, including bias arising from the randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of the reported result. An overall risk-of-bias judgment was then assigned for each study outcome based on these domain-level assessments. When published reports did not provide sufficient information to support

a judgment, the assessment was based on the available information, and attempts were planned to contact study investigators for clarification. RoB plot was created using the Risk-of-bias VISualization (robvis) tool.¹⁸

Synthesis Methods

A formal quantitative meta-analysis was not planned. Instead, findings were synthesized using a structured narrative approach. This decision was made a priori because the number of eligible studies was expected to be small, and heterogeneity was anticipated across study populations, HRT formulations and regimens, timing of initiation, follow-up duration, and reported outcomes. The evidence was therefore synthesized qualitatively.

Results

Study Selection

Study selection was conducted in accordance with the predefined eligibility criteria. Of the 1701 records initially identified, 311 duplicates were removed, and the remaining 1390 records underwent title and abstract screening. A total of 1231 records were excluded at this stage, and 159 reports were sought for full-text retrieval. Of these, 3 reports were not retrieved and 156 full-text reports were assessed for eligibility. Following full-text review, 150 reports were excluded, primarily because they used non-randomized study designs or did not include findings relevant to women aged 65 years and older. In total, 6 studies met the inclusion criteria and were included in the final review (Figure 1).

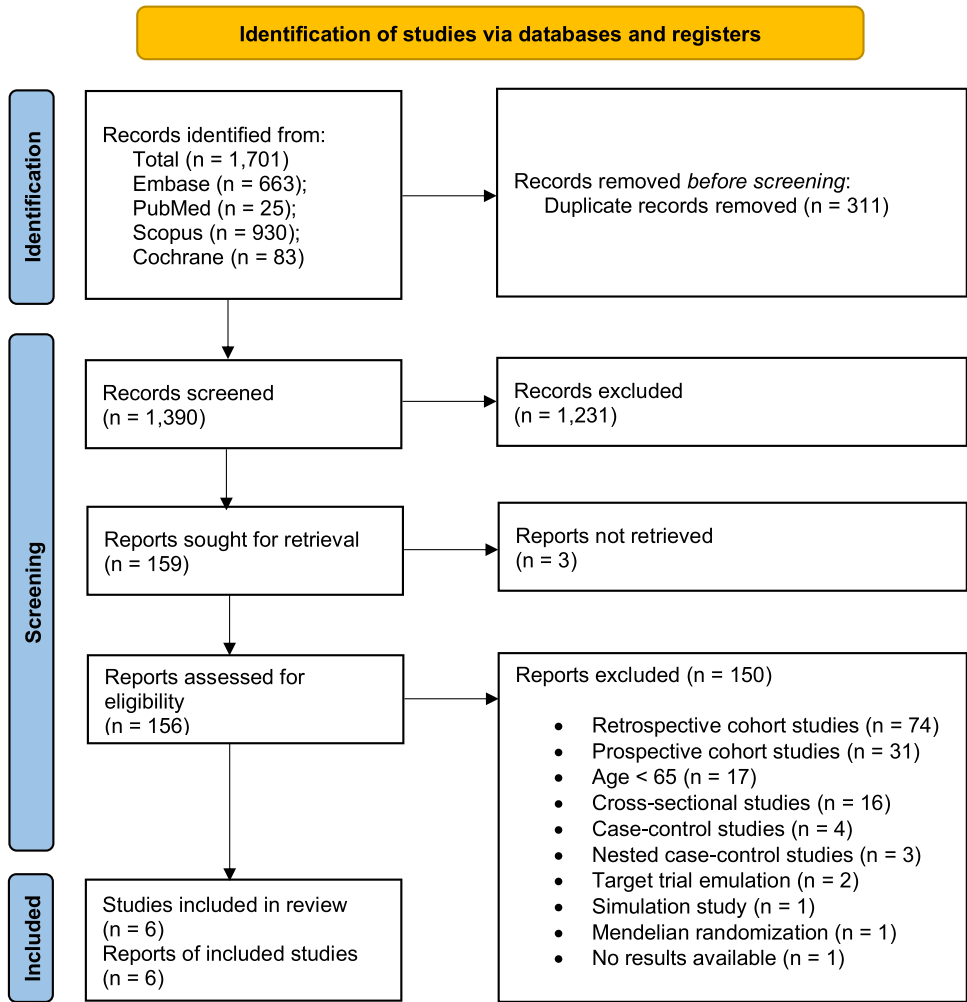


Figure 1 PRISMA flow diagram of study selection. The figure shows the process of identifying, screening, and assessing records for eligibility in the systematic review. **Abbreviation:** PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses.

Study Characteristics

Among the six eligible reports published between 2023 and 2026, all were secondary or post-trial analyses derived from three randomized placebo-controlled HRT trials. Three reports analyzed participants from the Women's Health Initiative Hormone Therapy Trials (WHI-HT), two analyzed participants from the Early versus Late Intervention Trial with Estradiol (ELITE), and one analyzed participants from the Kronos Early Estrogen Prevention Study (KEEPS). Thus, although six reports were included, the evidence was drawn from only three underlying trial populations, with WHI-HT participants contributing to three of the six reports. None of the reports presented outcomes specifically for women aged 65 years and older. The included reports varied in HRT regimen, trial setting, timing since menopause, and outcomes assessed, which included coronary heart disease risk, blood pressure, lipid profiles, glucose-related measures, insulin resistance, and subclinical vascular measures such as carotid intima-media thickness (CIMT). Most findings concerned intermediate cardiometabolic markers, while hard clinical outcomes were limited. No report evaluated renal outcomes, highlighting an important gap in the evidence available during the review period.

Women's Health Initiative Hormone Therapy Trials (WHI-HT)

Three studies were derived from the WHI-HT, in which postmenopausal women aged 50–79 years were randomized, depending on hysterectomy status, to oral conjugated equine estrogens (CEE) 0.625 mg/day alone or oral CEE 0.625 mg/day plus medroxyprogesterone acetate (MPA) 2.5 mg/day versus placebo. These WHI-HT analyses focused on coronary heart disease and thrombotic outcomes,¹⁹ blood pressure control among women with hypertension,²⁰ and longitudinal cardiovascular biomarker changes.²¹

In the WHI-HT study evaluating coronary heart disease risk, CEE alone was not significantly associated with increased risk compared with placebo at either 2 years (HR 1.15, 95% CI 0.78–1.71) or 5 years (HR 1.03, 95% CI 0.80–1.32), whereas CEE+MPA was associated with a significantly higher coronary heart disease risk at both 2 years (HR 1.49, 95% CI 1.05–2.11) and 5 years (HR 1.37, 95% CI 1.08–1.73), suggesting a less favorable cardiovascular profile for the combined regimen than for estrogen alone.¹⁹

In the WHI-HT biomarker analysis, both CEE and CEE+MPA lowered LDL-C relative to placebo, with ratios of geometric means of 0.89 (95% CI 0.88–0.91) and 0.88 (95% CI 0.86–0.89), respectively. Relative to placebo, HDL-C was 13.0% higher with CEE and 7.0% higher with CEE+MPA. Triglycerides were also 13.0% and 7.0% higher with CEE and CEE+MPA, respectively.²¹ Additional results were summarized in Table 1.

In the WHI-HT hypertension analysis, both active regimens were associated with modest but statistically significant elevations in systolic blood pressure compared with placebo. During the intervention phase, mean systolic blood pressure was 0.9 mm Hg higher with CEE alone (95% CI 0.2–1.5; $P = 0.02$) and 1.8 mm Hg higher with CEE+MPA (95% CI 1.2–2.5; $P < 0.001$). These findings were consistent during cumulative follow-up, with corresponding increases of 0.8 mm Hg (95% CI 0.1–1.4; $P = 0.02$) and 1.6 mm Hg (95% CI 1.0–2.3; $P < 0.001$), respectively.

ELITE

Two studies were based on ELITE, a single-center, randomized, double-blinded, placebo-controlled trial of healthy postmenopausal women stratified by time since menopause into early postmenopause (<6 years since menopause) and late postmenopause (≥ 10 years since menopause). Participants received oral micronized 17 β -estradiol 1 mg/day or placebo, with vaginal micronized progesterone gel 45 mg/day for 10 days each month added for women with an intact uterus. One ELITE analysis included 562 women (240 early and 322 late postmenopausal women) with the overall mean age of 60.5 years (SD 7.0). The results showed that HRT significantly increased total triglycerides, HDL-C, small LDL, large HDL, and triglyceride-to-cholesterol ratios in LDL and HDL, while decreasing LDL-C, total very-low-density lipoprotein (VLDL), small VLDL, intermediate-density lipoprotein (IDL), large LDL, and LDL peak diameter compared with placebo. Reported quantitative changes included 1.6 mg/dL higher triglyceride in LDL (95% CI 0.9 to 2.2; $P < 0.001$), 3.2 mg/dL lower cholesterol in LDL (95% CI -5.4 to -0.9 ; $P = 0.02$), 1.0 mg/dL higher triglyceride in HDL (95% CI 0.7 to 1.3; $P < 0.001$), 2.1 mg/dL higher cholesterol in HDL (95% CI 1.1 to 3.1; $P < 0.001$), and 0.8 Å lower LDL peak diameter (95% CI -1.3 to -0.4 ; $P < 0.001$).²² Another ELITE analysis included 535 women and found that associations between hormone levels and CIMT progression differed by time since menopause (interaction $P < 0.01$).

Table 1 Baseline Characteristics of Included Studies

First Author, Year	Trial Name and Study Design	Follow-Up Duration	Population	Intervention	Comparator	Outcomes	Main Results
Davis, 2026 ¹⁹	Secondary analysis of randomized, double-blind, placebo-controlled trials (WHI-HT trials)	Thrombotic outcomes were evaluated at 2 and 5 years; parent WHI trials had median intervention periods of 7.2 years for CEE-alone and 5.6 years for CEE+MPA	Postmenopausal women aged 50–79 years; 10,739 women with prior hysterectomy were randomized in the CEE-alone trial and 16,608 women with an intact uterus in the CEE+MPA trial	Oral CEE 0.625 mg/day alone or oral CEE 0.625 mg/day plus MPA 2.5 mg/day	Placebo	Coronary heart disease, stroke, venous thromboembolism, and major cardiovascular disease	Baseline statin or aspirin use did not significantly modify the effects of oral HT on arterial or venous thrombotic outcomes at 2 or 5 years; findings may have been underpowered because baseline statin and aspirin use were infrequent.
Jiang, 2023 ²⁰	Secondary analysis of randomized, double-blind, placebo-controlled trials (WHI-HT trials)	Blood pressure was measured at baseline and up to 10 annual follow-up visits during the trial; intervention effect was estimated through year 6, with cumulative follow-up extending to a median of 13 and 16 years after randomization	Subsample of 9332 postmenopausal women with hypertension at baseline from the WHI hormone therapy trials	Oral CEE 0.625 mg/day alone or oral CEE 0.625 mg/day plus MPA 2.5 mg/day	Placebo	Systolic blood pressure and number of antihypertensive medication classes used during follow-up	Compared with placebo, both CEE-alone and CEE +MPA were associated with small but statistically significant increases in systolic blood pressure during intervention and cumulative follow-up, without differences in antihypertensive medication use over time.
Nudy, 2025 ²¹	Secondary analysis of randomized, double-blind, placebo-controlled trials (WHI-HT trials)	Biomarkers were measured at baseline and at 1, 3, and 6 years during the intervention phase	Subsample of WHI hormone therapy trial participants: 1188 women from the CEE-alone trial and 1508 women from the CEE+MPA trial; postmenopausal women aged 50–79 years at randomization	Oral CEE 0.625 mg/day alone or oral CEE 0.625 mg/day plus continuous MPA 2.5 mg/day	Placebo	LDL-C, HDL-C, triglycerides, total cholesterol, lipoprotein(a), glucose, insulin, and HOMA-IR	Compared with placebo, HT lowered LDL-C, lipoprotein(a), and HOMA-IR and increased HDL-C, but increased triglycerides in both the CEE-alone and CEE+MPA trials.

Sriprasert, 2025 ²²	Randomized, double-blind, placebo-controlled trial stratified by time since menopause (ELITE)	Median follow-up 4.8 years (range 0.5–6.7); fasting blood samples were obtained every 6 months	562 healthy postmenopausal women: 240 early post-menopause and 322 late postmenopausal women	Oral micronized 17 β -estradiol 1 mg/day; women with a uterus also received vaginal micronized progesterone gel 45 mg/day for 10 days each month	Placebo	Standard lipids and lipoprotein particle subfractions; associations with atherosclerosis progression	HT increased triglycerides and HDL-C and decreased LDL-C, VLDL, IDL, and large LDL; effects were similar in early and late post-menopause, with no significant interaction by time since menopause.
Chen, 2025 ²³	Post-trial analysis of the randomized, double-blind, placebo-controlled ELITE trial stratified by time since menopause	Median trial follow-up was 4.8 years (range 0.5–6.7); sex steroid hormones and SHBG were measured at baseline, 12 months, and 36 months, and CIMT was assessed every 6 months	535 healthy postmenopausal women with follow-up sex steroid hormone, SHBG, and CIMT data; early post-menopause <6 years and late post-menopause $\geq$ 10 years since menopause	Oral micronized estradiol 1 mg/day; women with a uterus also received vaginal micronized progesterone gel 4% (45 mg/day) for 10 days each month	Placebo	Serum estradiol, estrone, testosterone, SHBG, and CIMT progression	Associations with CIMT progression differed by time since menopause: in early post-menopause, higher SHBG was inversely associated with CIMT progression, whereas in late post-menopause, higher estradiol, estrone, and SHBG were positively associated with CIMT progression.
Kantarci, 2024 ²⁴	Continuation study of the randomized, double-blind, placebo-controlled KEEPS trial	Participants received treatment for 4 years in KEEPS and were re-evaluated approximately 14 years after randomization and 10 years after trial completion	Recently postmenopausal women originally enrolled in KEEPS (n=727); the continuation study included 299 participants (41%) with a mean age of 67 years	Oral conjugated equine estrogens 0.45 mg/day or transdermal 17 β -estradiol 50 μ g/day; both active groups also received oral micronized progesterone 200 mg/day for 12 days each month	Placebo	Blood pressure, waist-to-hip ratio, fasting glucose, insulin, HOMA-IR, lipid profile, self-reported diabetes, and diabetes medication use	Cardiometabolic measures were generally similar across groups at 14-year follow-up, with no clear long-term cardiovascular or metabolic benefit or harm from prior hormone therapy; although self-reported diabetes and diabetes medication use were more frequent in the placebo group, laboratory measures did not support these differences.

Abbreviations: CEE, conjugated equine estrogens; CHD, coronary heart disease; CIMT, carotid artery intima-media thickness; ELITE, Early versus Late Intervention Trial with Estradiol; HDL-C, high-density lipoprotein cholesterol; HOMA-IR, homeostatic model assessment for insulin resistance; HT, hormone therapy; IDL, intermediate-density lipoprotein; KEEPS, Kronos Early Estrogen Prevention Study; LDL-C, low-density lipoprotein cholesterol; MPA, medroxyprogesterone acetate; SHBG, sex hormone-binding globulin; TG, triglycerides; VLDL, very low-density lipoprotein; VTE, venous thromboembolism; WHI, Women's Health Initiative; WHI-HT, Women's Health Initiative hormone therapy.

Among late postmenopausal women, CIMT progression was positively associated with estradiol [$\beta = 1.93 \mu\text{m}/\text{year}$, 95% CI 0.58 to 3.29; $P = 0.005$], estrone [$\beta = 2.36$, 95% CI 1.12 to 3.61; $P < 0.001$], and sex hormone-binding globulin (SHBG)[$\beta = 1.25$, 95% CI 0.07 to 2.43; $P = 0.037$], whereas among early postmenopausal women, CIMT progression was inversely associated with SHBG [$\beta = -1.80$, 95% CI -3.36 to -0.24 ; $P = 0.024$], while associations with estradiol and estrone were inverse but not significant.²³

KEEPS

The remaining study came from the KEEPS continuation cohort, which followed women originally randomized within 6 to 36 months after menopause to oral CEE 0.45 mg/day, transdermal 17 β -estradiol 50 $\mu\text{g}/\text{day}$, or placebo, with oral micronized progesterone 200 mg/day for 12 days each month given in the active-treatment groups. Of the 727 women in the original KEEPS trial, 299 (41%) participated in the continuation study. At follow-up, participants had a mean age of 67 years (range 58–73 years) and were assessed approximately 14 years after randomization and 10 years after completion of the original 4-year trial. The KEEPS continuation study found no statistically significant between-group differences in systolic or diastolic blood pressure, waist-to-hip ratio, fasting glucose, insulin, HDL-C, LDL-C, total cholesterol, triglycerides, or homeostatic model assessment of insulin resistance (HOMA-IR) at the continuation visit, and no significant differences were observed for change from baseline to year 14 (all $P > 0.05$). The only significant between-group differences were for self-reported diabetes and diabetes medication use, both of which were more common in the placebo group. Specifically, self-reported diabetes occurred in 6% of placebo participants versus 0% in both the oral CEE and transdermal estradiol groups ($P = 0.007$), and diabetes medication use was reported by 8% of placebo participants versus 1% in each active-treatment group ($P = 0.045$ for placebo vs oral CEE; $P = 0.02$ for placebo vs transdermal estradiol). However, these differences were not supported by laboratory measures of glycemia or insulin resistance.²⁴

RoB Assessment

RoB was assessed for 5 out of 6 studies. Notably, the Chen (2025) study²³ was not assessed using RoB 2 because the included result was based on a post-trial observational analysis of the association between on-trial hormone levels and CIMT progression, rather than the effect of randomized assignment to HRT versus placebo. Of the remaining 5 studies, RoB was judged as some concerns for 4 studies and high risk of bias for 1 study. As shown in Figure 2, the most frequent source of concern was selection of the reported result, mainly because prespecified analysis plans were not available and several papers reported multiple eligible outcomes, time points, or exploratory analyses. One study had high risk of bias because only a minority of the original randomized cohort contributed follow-up outcome data, raising concern about

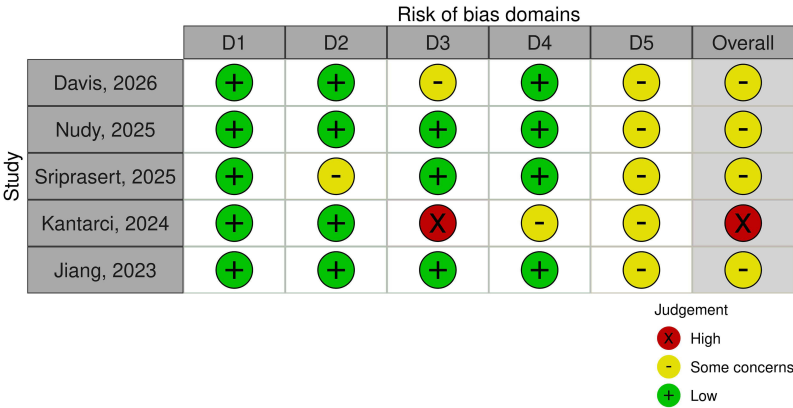


Figure 2 Risk-of-bias assessment of included reports using the RoB 2 tool. The figure summarizes domain-level and overall risk-of-bias judgments for reports that evaluated outcomes based on randomized hormone therapy assignment: Davis et al,¹⁹ Nudy et al,²¹ Sriprasert et al,²² Kantarci et al,²⁴ and Jiang et al²⁰ D1 indicates bias arising from the randomization process; D2, bias due to deviations from intended interventions; D3, bias due to missing outcome data; D4, bias in measurement of the outcome; and D5, bias in selection of the reported result. Green indicates low risk of bias, yellow indicates some concerns, and red indicates high risk of bias. Chen et al²³ was not assessed using RoB 2 because the included analysis examined associations between on-trial hormone levels and carotid intima-media thickness progression rather than the effect of randomized assignment to hormone replacement therapy versus placebo.

bias due to missing outcome data. Other domains, including the randomization process, deviations from intended interventions, and outcome measurement, were generally judged as low risk of bias across studies.

Discussion

This systematic review demonstrated that HRT has mixed cardiometabolic effects in older postmenopausal women and that these effects differ by regimen, outcome, and timing of use. However, these conclusions are based on a limited evidence base comprising only six secondary or post-trial analyses derived from three parent trials—the WHI-HT, ELITE, and KEEPS. This heavy reliance on a small number of trial populations limits the independence, generalizability, and certainty of the available evidence.

In the included WHI-HT analyses, both oral CEE alone and oral CEE plus MPA improved some intermediate markers, including LDL-C and lipoprotein(a), and lowered HOMA-IR, but both regimens also increased triglycerides. In parallel, both regimens were associated with small but statistically significant increases in systolic blood pressure. These findings indicate that favorable changes in one metabolic domain can occur at the same time as less favorable changes in another, and they help explain why improvement in selected biomarkers cannot be assumed to reflect overall cardiovascular benefit.

This interpretation is also consistent with earlier trial evidence. The Postmenopausal Estrogen/Progestin Interventions (PEPI) trial, which enrolled relatively younger postmenopausal women aged 45–64 years, reported favorable effects of estrogen-containing regimens on several lipid measures, especially lower LDL-C and higher HDL-C, which initially supported the idea that HRT might reduce cardiovascular risk.²⁵ However, Heart and Estrogen/progestin Replacement Study (HERS) which focused on older women with established coronary disease aged 44–79 years, did not show a reduction in coronary events with estrogen plus progestin.²⁶ The Cochrane review later pooled trials with a broader age distribution, which concluded that HRT should not be used to prevent cardiovascular disease because it does not provide overall cardiovascular protection and increases some harms, including stroke and venous thromboembolism.²⁷

Another important finding of this review is that the effect of therapy should not be discussed as if all regimens are the same. In the WHI-HT evidence, CEE alone was not clearly associated with higher coronary heart disease risk at 2 or 5 years, whereas CEE plus MPA showed a less favorable coronary pattern. That distinction is important because estrogen-only therapy and combined estrogen–progestin therapy are often grouped together in broad discussions of HRT, even though the cardiovascular findings are not the same. Recent evidence outside the trials included here also supports the need to separate regimens and route of administration rather than treating HRT as a single exposure. A large contemporary target-trial emulation from Sweden found that oral estrogen–progestin therapy was associated with higher risks of heart disease and venous thromboembolism, whereas transdermal regimens showed a different risk pattern.²⁸ This is especially relevant for future research in older women, in whom oral and transdermal regimens may have different clinical implications.

The ELITE-based studies added another important point about timing hypothesis. In the lipid subfraction analysis included in this review, oral estradiol changed lipids in broadly similar ways in early and late postmenopause. By contrast, the vascular findings were not the same across timing groups. In the ELITE-based study of sex steroid hormones and carotid intima-media thickness (CIMT) progression, higher estradiol, estrone, and SHBG were associated with faster CIMT progression in late postmenopause, whereas in early postmenopause higher SHBG was associated with slower progression. These findings fit with the original ELITE trial, which showed that oral estradiol slowed CIMT progression when started within 6 years of menopause, but not when started 10 or more years after menopause.²⁹ Together, these studies suggest that hormone exposure may continue to affect metabolic markers in later postmenopause, but that the vascular response may be less favorable once women are further from menopause.

The KEEPS continuation study also provided a useful contrast. In that study, there were no clear long-term differences across groups in blood pressure, glucose-related measures, insulin resistance, or lipid levels at follow-up. However, that result should be read in light of the study population. KEEPS enrolled recently menopausal women who were selected for good cardiovascular health at baseline, and the continuation cohort included only 299 of the original 727 participants. These features limit how far the results can be applied to women who initiate therapy later in life or who

already have a higher cardiometabolic burden. The neutral long-term findings from KEEPS therefore should not be interpreted as direct evidence for safety of late initiation in women aged 65 years and older.

A clinically important message from this systematic review is that decisions in older women should not be driven by a single favorable biomarker. Lower LDL-C after treatment may appear reassuring, but that finding does not remove concern about higher triglycerides, higher blood pressure, or a less favorable thrombotic profile. This is particularly relevant in older women, who are more likely to have hypertension, obesity, diabetes, or existing vascular disease. In these settings, small adverse effects across several pathways may matter even when one or two laboratory measures improve. Current guidance reflects this concern by advising against the use of HRT for the primary prevention of chronic conditions in asymptomatic postmenopausal women and by emphasizing individualized treatment based on symptom burden, baseline risk, and the specific product used.³⁰

From an evidence and research perspective, this review highlights a substantial gap between the evidence currently available and the evidence clinicians and patients need. First, most included studies evaluated intermediate outcomes such as lipid levels, blood pressure, insulin resistance, or CIMT. Although these measures are informative, they do not directly answer the questions regarding hard clinical outcomes, namely whether initiation or continuation of HRT after age 65 affects the risks of myocardial infarction, stroke, venous thromboembolism, diabetes, kidney outcomes, hospitalization, or treatment discontinuation. Future studies should also incorporate hemostatic outcomes, including coagulation parameters and platelet function testing, to clarify the mechanisms underlying the thrombotic risks associated with specific HRT regimens. These measures should be assessed alongside clinical thrombotic outcomes and evaluated across different formulations, doses, and routes of administration. Second, women aged 65 years and older remain underrepresented in recent clinical trials. Although this review aimed to focus specifically on this population, most of the included evidence was derived from broader postmenopausal cohorts rather than from trials explicitly designed to evaluate late initiation or continued use after age 65. Third, renal outcomes have remained largely unexamined in HRT trials and secondary analyses published over the past five years, representing an important gap, particularly in older women who may already have age-related decline in kidney function or other chronic conditions that increase renal vulnerability. Fourth, direct comparisons across hormone formulations, doses, and routes of administration remain limited in women with overlapping cardiovascular-kidney-metabolic conditions, despite the likelihood that these treatment characteristics influence safety and tolerability. Fifth, the available evidence is still heavily derived from secondary and post-trial analyses rather than from trials prospectively designed to evaluate cardiometabolic or renal outcomes in older women, which limits both the strength and the clinical specificity of the conclusions that can be drawn. Sixth, the current literature provides limited evidence on how concomitant medications may modify the effects of HRT, even though these factors are central to individualized decision-making in routine care.

While this systematic review identifies important gaps in the current evidence base, several limitations of the review should be acknowledged when interpreting these findings. The number of eligible studies was small. Most included studies were secondary or post-trial analyses rather than primary randomized studies designed for older women. There was substantial heterogeneity in regimen, population, timing since menopause, follow-up length, and outcome selection. Several studies examined multiple outcomes and time points, and prespecified analysis plans were not always available. Finally, although study-level risk of bias was assessed using RoB 2, a formal outcome-level certainty-of-evidence assessment using GRADE was not prespecified or conducted, which limits the ability to characterize confidence in the findings systematically.

Conclusions

The recent randomized evidence relevant to cardiometabolic outcomes of HRT in older postmenopausal women is sparse, indirect, and heavily concentrated within secondary or post-trial analyses of three parent trial programs. Although these analyses identified both favorable and unfavorable changes across intermediate metabolic and cardiovascular measures, they do not provide direct evidence to determine the overall cardiovascular safety or benefit of initiating HRT after age 65 years. Evidence was further limited by participant overlap, heterogeneity of regimens and outcomes, reliance on surrogate measures, risk-of-bias concerns, and underrepresentation of women initiating treatment at older ages. No

included report presented renal outcomes. These findings underscore the need for future clinical trials and cohort studies evaluating major cardiovascular, metabolic, and kidney outcomes in older postmenopausal women.

Abbreviations

CENTRAL, Cochrane Central Register of Controlled Trials; CEE, Conjugated equine estrogens; CIMT, Carotid intima-media thickness; ELITE, Early versus Late Intervention Trial with Estradiol; HDL-C, High-density lipoprotein cholesterol; HERS, Heart and Estrogen/progestin Replacement Study; HOMA-IR, Homeostatic model assessment of insulin resistance; HRT, Hormone replacement therapy; IDL, Intermediate-density lipoprotein; KEEPS, Kronos Early Estrogen Prevention Study; LDL-C, Low-density lipoprotein cholesterol; MPA, Medroxyprogesterone acetate; NAMS, North American Menopause Society; PECO, Population, Exposure, Comparator, Outcome; PEPI, Postmenopausal Estrogen/Progestin Interventions trial; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-Analyses; PROSPERO, International Prospective Register of Systematic Reviews; RCT, Randomized controlled trial; RoB, Risk of bias; SD, Standard deviation; SHBG, Sex hormone-binding globulin; USFDA, US Food and Drug Administration; VLDL, Very-low-density lipoprotein; WHI-HT, Women's Health Initiative Hormone Therapy Trials.

Author Contributions

All authors made a significant contribution to the work reported, whether that is in the conception, study design, execution, acquisition of data, analysis and interpretation, or in all these areas; took part in drafting, revising or critically reviewing the article; gave final approval of the version to be published; have agreed on the journal to which the article has been submitted; and agree to be accountable for all aspects of the work.

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